# DATA COMMUNICATION FINAL T1.2023

***Q1 : Filling the approciate word(s) from the box into the following two charts :***

***( analog , sin waves , non periodic , periodic , composite , simple , analog , sine wave )***

1) **Singal :** ***analog & sin waves & composite & non periodic***
2) **Singal :** ***analog & sine wave & simple & periodic***

***Q2 : Choose the correct answer :***

**1) Radio waves can penetrate walls, this property can considered as :**
   a) Advantage
   b) Disadvantage
   c) Advantage & Disadvantage

**2) Oscillation of sin wave is associated with its :**
   a) Cycle
   b) Amplitude
   c) Phase

**3) Which of the following is independent of the medium :**
   a) Frequency
   b) Wavelength
   c) Speed

**4) In a human voice :**
   a) The number of frequencies is infinite , but the range is limted
   b) The number of frequencies is finite , but the range is limted
   c) The number of frequencies is infinite , but the range is umlimted

**5) Bit length is measured by :**
   a) Bits
   b) Bits/second
   c) Meters

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6) **The main difference between periodic abd non-periodic digital signals is :**
   a) Bandwidth
   b) Frequencies
   c) Amplitude

7) **A block of 256 sequential 12-bit data word is transmitted serially in 0.016s, Calculate the time duration of 1 word :**
   a) 1.33 µs
   b) 62.5 µs
   c) 16 µs

8) **A PCM transmission requires :**
   a) More bandiwdth than AM
   b) Less bandiwdth the AM
   c) Same bandiwdth as AM

9) **A bandwidth of :**
   a) 10 KHZ for transmission of speech signals
   b) 200 KHZ for transmission of speech signals
   c) 4 KHZ for transmission of speech signals

10) **UHF propagation takes place primarily through :**
   a) Ground wave
   b) Sky wave
   c) Line-Of-Sight wave

11) **Indicate which of the following system is digital :**
   a) PPM
   b) PCM
   c) PWM

12) **A signal of 140 MHZ has a bandwidth of s a bandwidth ±20 MHZ, what is the Nyquist sampling rate :**
   a) 80 MHZ
   b) 160 MHZ
   c) 40 MHZ
   d) 320 MHZ

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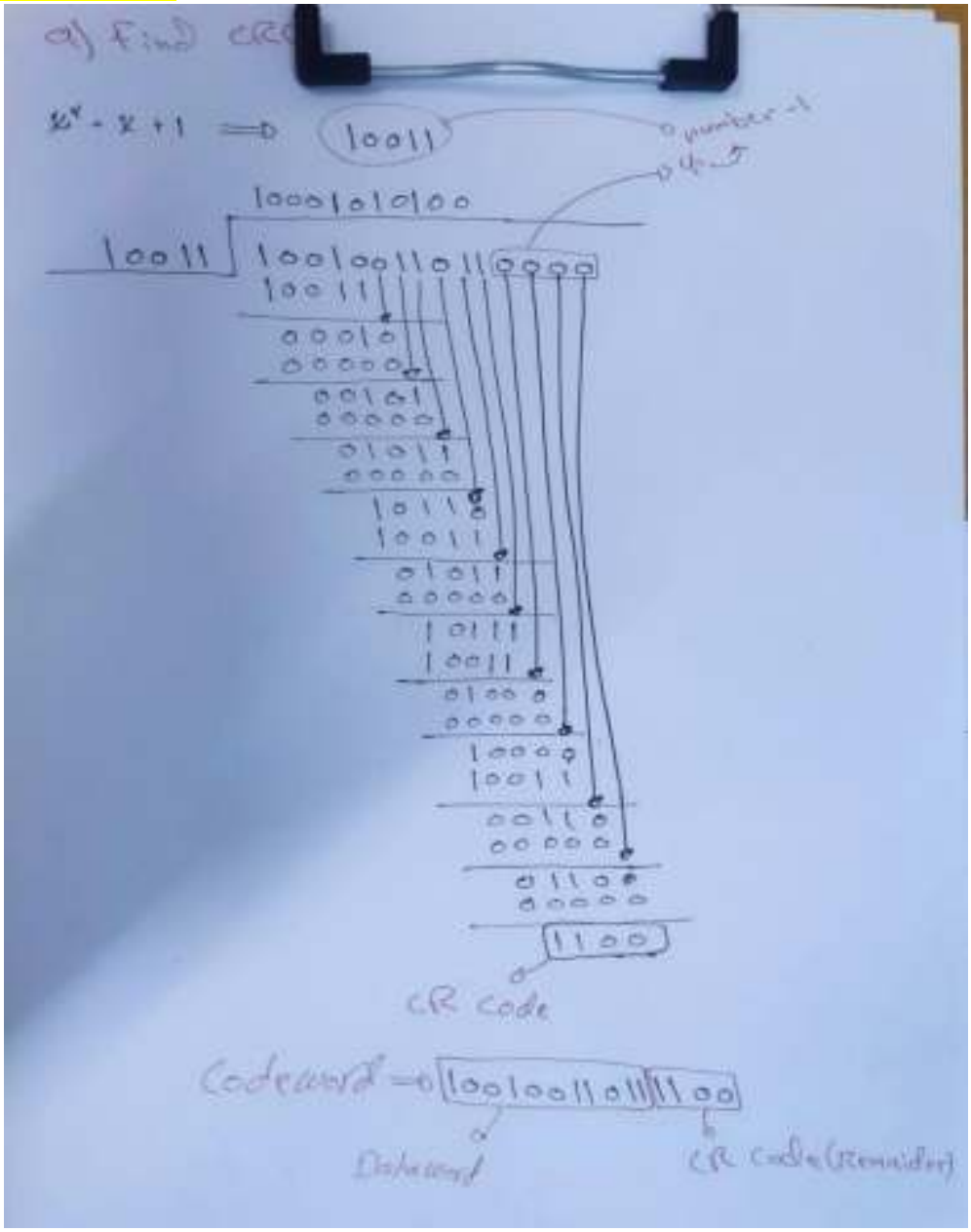
***Q3 ( 10 marks ) : If the data combination is represented as 10010011011 , Which will be augmented according to the generator ( $x^4 + x + 1$ )***

***a) Find CRC .***

***b) If the received combination is ( $x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^3 + x^2$ ) , what is the decoding result for this combination .***

***SOLUTION :***

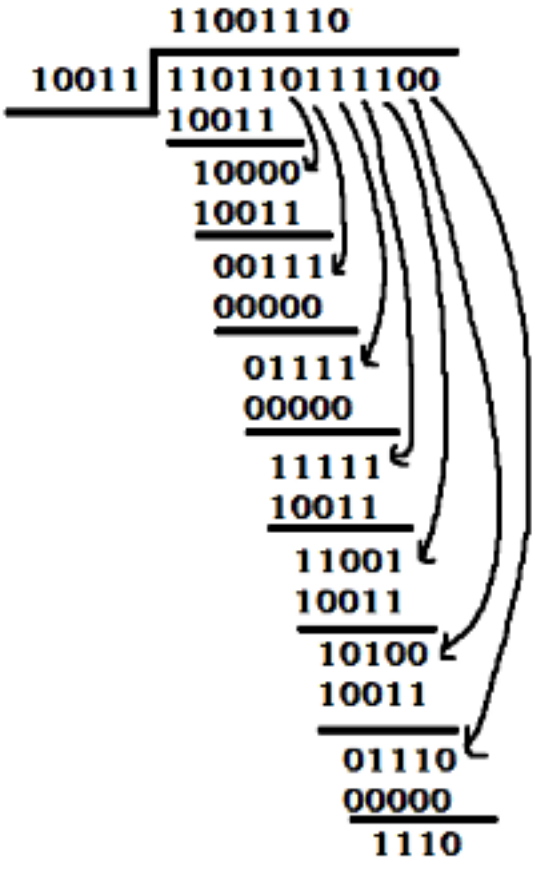
***a)***




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***b)***

**AT RECEIVER ( DECODING ) :**




**NOT ZERO , NOT ACCEPT**

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***Q4 ( 11 marks ) : For 11 binary bit string 01001100011 :***

***a) Find the number of transistions for the encoding schemes Manchester & Diffrential Manchester .***

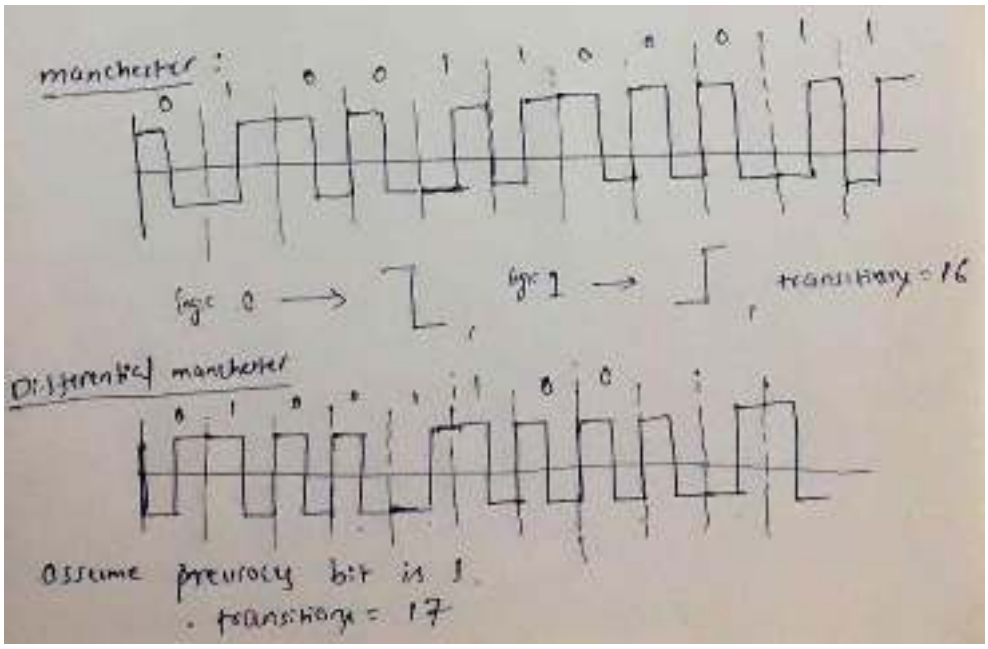
***b) If the above combination rate is 10Mbps, Calculate the average signal rate using Manchester code .***

***c) Calculate the number of data elements carried by each signal element in Diffrential Manchester code .***

***d) Calculate maximum bandwidth .***

***SOLUTION :***

***a-***




***b- ( S = 10M baud rate )***

***c- ( r = ½ )***

***d- ( Bmin = 10MHZ )***

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- ***More Question On Data Communication & Networking***

***Q1 : $F(t)=( 12\cos(4t) )^2$ , using identity :-***
***$\cos x * \cos y = 0.5 * ( \cos(x-y) + \cos(x+y) )$***

***a) Find Period .***
***b) Find fs .***

***a- $144 * 0.5 ( \cos(4t-4t) + \cos( 4t+4t ) )$***

***$72 ( 1 + \cos(8t))$***

***$72 + 72 \cos(8t)$***

***$T = 2\pi/8$***

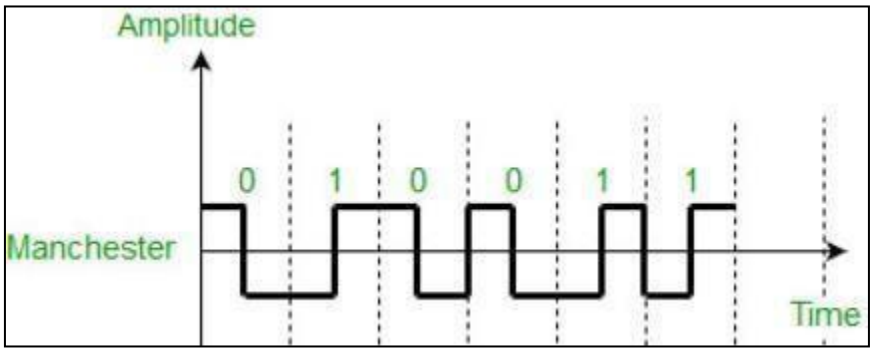
***a) $Fmax = 8/2\pi$***

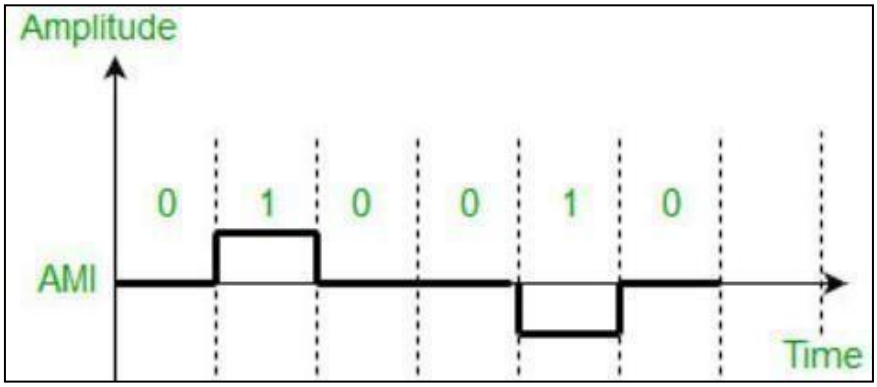
***$Fs = 2 * Fmax = 8/\pi$***

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- ***More Question On Data Communication & Networking***

## *Q2 : For this two figure shown below :*




***a) Find bit per signal for each figure .***

***( (½) Fig.1 , (1) Fig.2 )***

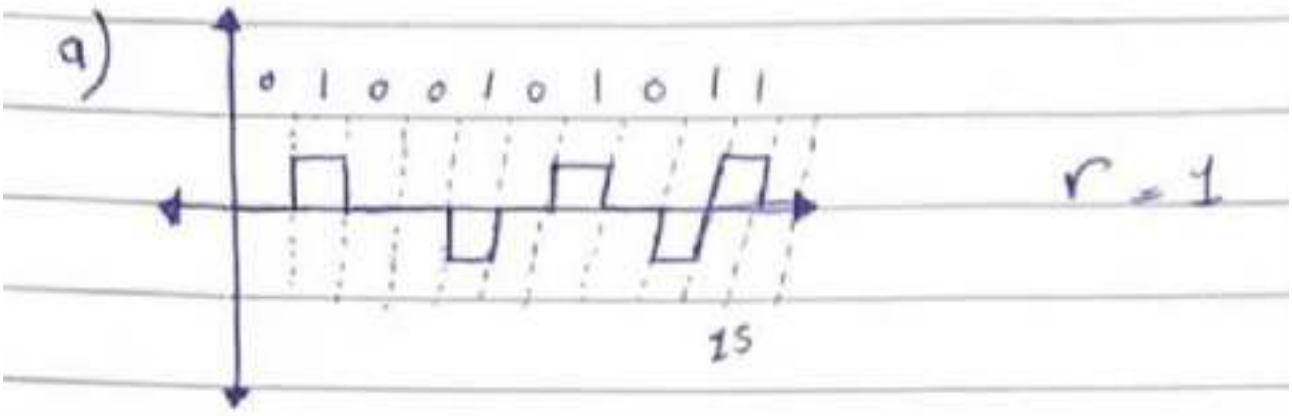
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- ***More Question On Data Communication & Networking***

## *Q3 : Consider the following sequence : ( 0100101011 )*

a) ***Represent the above sequence into bipolar-AMI waveform .***
b) ***Calculate the bit rate if the sequence is sent in one second .***
c) ***Calculate the baud rate of the transmitted sequence .***
d) ***Sketch the above bit stream into Manchester-Code .***
e) ***With respect to above results calculate minimum required bandwidth .***
f) ***Consider SNR=24 DB , Calculate the maximum channel capacity & Number of level FOR AMI .***

***a)***

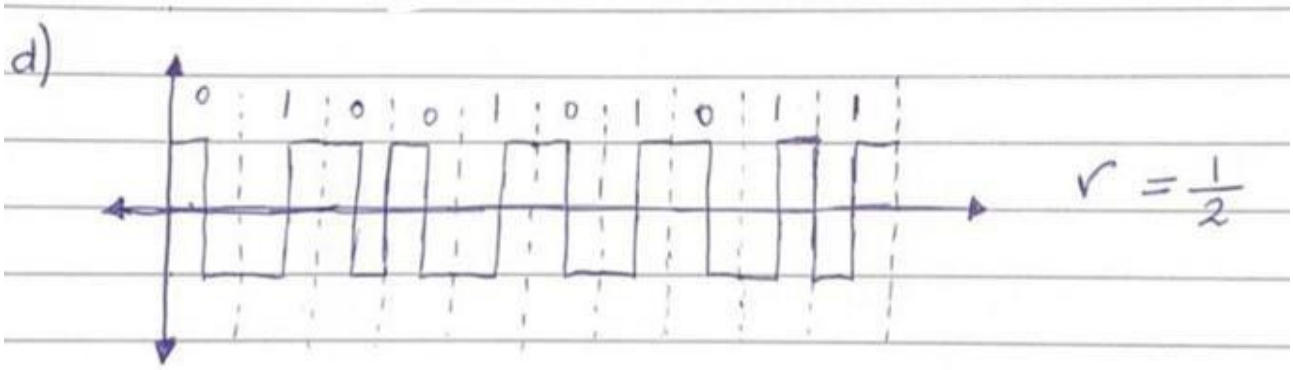



***b) N = 10 bps***

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*c)* ***S = N = 10 baud rate***

*d)*




*e)* ***B(min) [AMI] = S(avg) = ½ * N * 1/r***
***= ½ * 10 * 1/1 = 5 HZ***

***B(min) [Manchester] = S(avg) = ½ * N * 1/r***
***= ½ * 10 * 1/½ = 10 HZ***

*f)* ***SNR(db) = 24 db ,, SNR(db) = 10 Log( SNR )***

***24 = 10 Log ( SNR ) ➔ SNR = 251.2***

***** C = B * $Log_2$( 1 + SNR )***

***C = 5 * $Log_2$( 1 + 251.2 ) ➔ C = 39.89 bps***

***** NyquistRate = 2 * B * $Log_2$( L )***

***39 = 2 * 5 * $Log_2$( L ) ➔ L = 14.9***

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